

Name (print): _____

MYUH# _____

In-class Quiz
#01: Numbering Systems

1. Convert the unsigned 8-bit binary number to decimal; convert the decimal number to unsigned 8-bit binary
1) 0b 1001 1100; 2) 73;

2. Convert the hexadecimal number (unsigned) to decimal; convert the decimal number to hexadecimal
1) 0x39; 2) 527;

3. Convert the binary number to hexadecimal; convert the hexadecimal number to binary
1) 0b 0111 1010 0101 0011; 2) 0x3AE2.

4. Convert the numbers into the specified number system. (use 0-9 and A-J to represent the digit from 0 to 19. For example, you can use A to represent the digit 10, and use G to represent the digit 16):
 - 1) Convert $(273)_8$ to base-6 (Radix = 6);
 - 2) Convert $(A3)_{14}$ to base-16 (Radix = 16).

5. Change the signed 8-bit binary, using two's complement representation, back to decimal:
1) 0b 1011 0100; 2) 0b 0111 0001;

6. For the following questions,
step 1) convert the decimal numbers into signed binary using 2's complement representation,
step 2) perform the binary addition,
step 3) then convert the results back to decimal.

Hint: To add two signed 8-bit binary numbers using two's complement, simply align the values on the least significant bit and add, propagating any carry to the bit one position high. Assume all results wrap around using 8-bit two's complement. (If you have a carry past the end of the word, just drop it.)

- 1) $(-32)+17$; 2) $(-28)+(-43)$; 3) $(-52)+89$